

Computer Networks

Lecture 26 Network Layer

Content



- **Distance Vector Routing**

Distance Vector Routing : Sharing

- Routers share routing information with their neighbors.
- A router can learn about unknown networks through neighbors
Example: A learns about E via C, and C learns about D via A
- Neighbors help each other improve routing tables

- Each router sends its routing table to neighbors
- Only first two columns are shared:
 - Destination
 - Distance (cost)

The next hop is not shared, because: Neighbor will replace it with the sender's address

Distance Vector Routing : Update

When a router receives a 2-column table (destination, cost) from a neighbor, it updates its own table.

- Step 1: Add the link cost (distance to neighbor) to each received entry
New cost = (neighbor's cost to destination) + (cost to reach neighbor)
- Step 2: For any usable entry,
set next hop = sender (neighbor) Because the route now goes via that neighbor
- Step 3: Compare with Existing Table For each destination:
 - Case 1: Different Next Hop Choose the route with lower cost
⇒ If costs are equal → keep the old route
 - Case 2: Same Next Hop Always update with new value Even if new cost is higher or infinity
Because it reflects the latest network condition
⇒ Old route may no longer exist

Distance Vector Routing : Sharing

When does a node send its partial routing table (only two columns) to all its immediate neighbors?

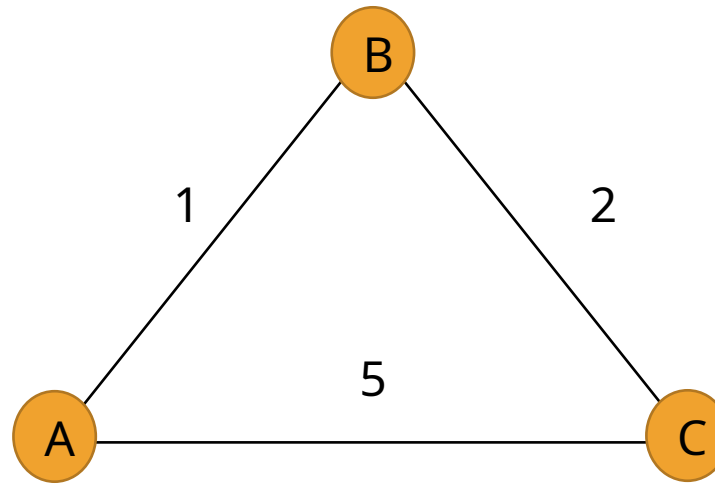
Periodic Update A node sends its routing table, normally every 30 s, in a periodic update. The period depends on the protocol that is using distance vector routing.

Triggered Update A node sends its two-column routing table to its neighbors anytime there is a change in its routing table. The change can result from the following.

- A node receives a table from a neighbor, resulting in changes in its own table after updating.
- A node detects some failure in the neighboring links which results in a distance change to infinity.

Question

Q1: Find routing tables of routers A, B and C using distance vector routing algorithm



Question

Q2: Consider following are the routing table of Router B , C and F. The measure delays of B,C and F via A are (1,2,3). Calculate the final vector of A for a new packet.

Q3: If the packet comes to A then it has to go *F* via which path it will go?

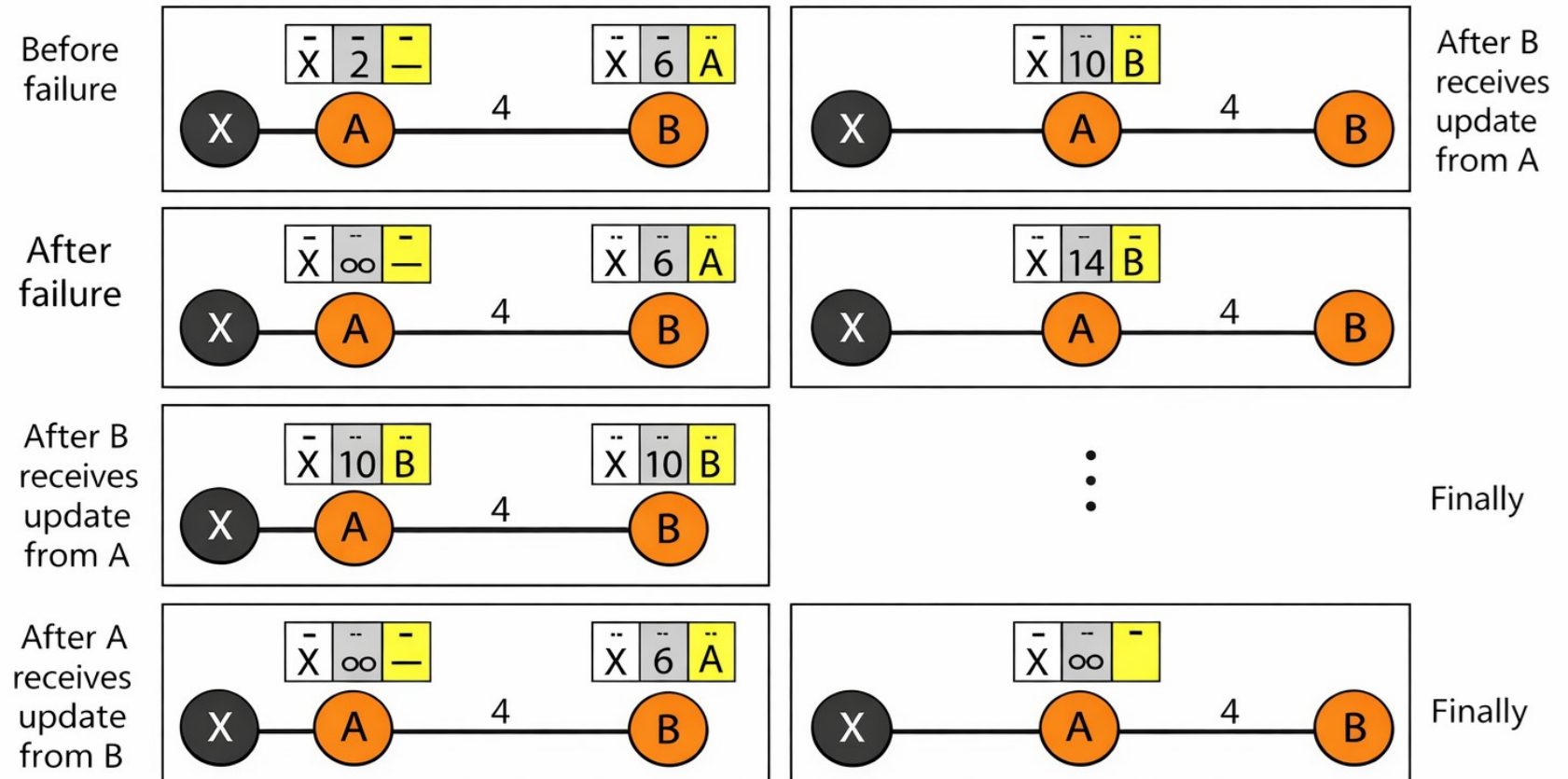
Q4: If the packet comes to A then it has to go *E* via which path it will go?

Routers	B	C	F
A	1	2	3
B	0	3	1
C	1	0	4
D	4	1	6
E	3	2	1
F	5	1	0

Q5: Due to change in the network B has received the table's of A and E (4,3,6,8,0,9) The measures delay of E is 3 from B. Calculate the routing table for router B.

Two-Node Loop Instability

A problem with distance vector routing is instability, which means that a network using this protocol can become unstable.



Two-Node Loop Instability

Possible Solutions

1. Defining Infinity: The first obvious solution is to redefine infinity to a smaller number, such as 100.
2. Split Horizon: In this strategy, instead of flooding the table through each interface, each node sends only part of its table through each interface.

Example:

If, according to its table, node B thinks that the optimum route to reach X is via A, it does not need to advertise this piece of information to A; the information has come from A (A already knows).

Two-Node Loop Instability

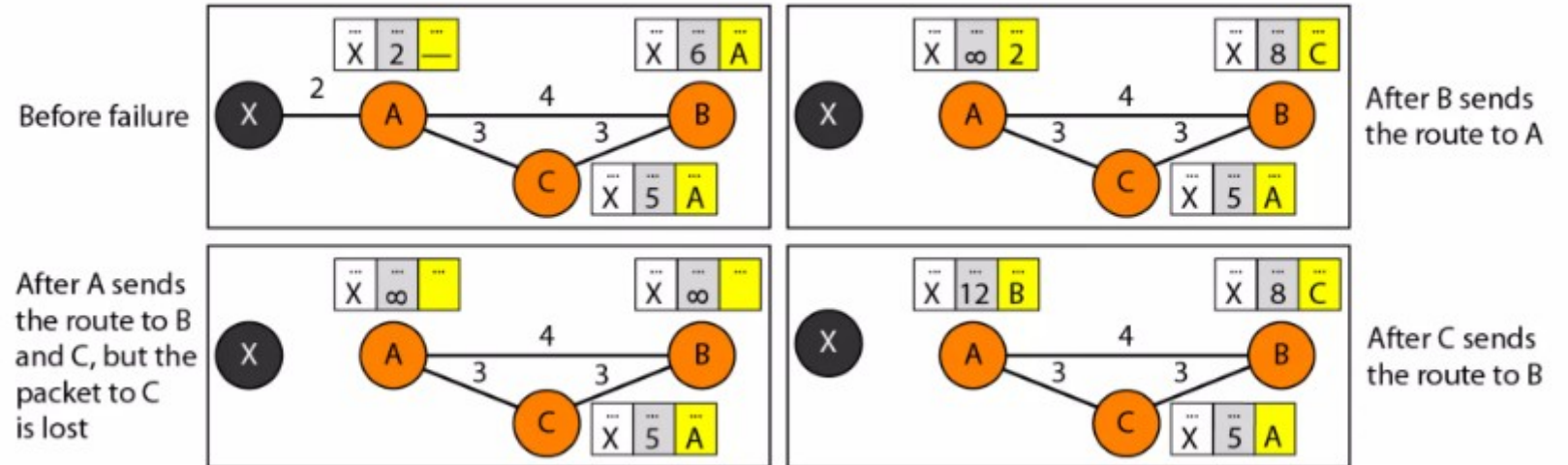
Possible Solutions

3. Split Horizon and Poison Reverse:

Split horizon has a drawback: when a node stops advertising a route, neighbors cannot tell whether the route is unavailable or just suppressed due to split horizon. To solve this, poison reverse is used—the node still advertises the route but marks it with infinite distance, indicating the information came from that neighbor and should not be used.

Three-Node Loop Instability

The two-node instability can be avoided by using the split horizon strategy combined with poison reverse. However, if the instability is between three nodes, stability cannot be guaranteed.



Question

Q6. Consider a network with three nodes A , B , and X . The link costs are as follows:

- $A \leftrightarrow B = 1$
- $B \leftrightarrow X = 1$
- A has no direct link to X

Initially, the routing tables are stable.

1. Construct the initial routing tables for nodes A and B with respect to destination X .
2. Suppose the link between B and X fails.
3. Show the step-by-step updates of routing tables at nodes A and B for at least 3 iterations using the distance vector algorithm.
4. Explain how the count-to-infinity problem occurs in this case.

Question

Q7. Consider a network with four nodes A, B, C, and X arranged as follows:

- $A \leftrightarrow B = 1$
- $B \leftrightarrow C = 1$
- $C \leftrightarrow X = 1$

Initially, all routing tables are stable.

1. Determine the initial shortest path cost from nodes A, B, and C to destination X.
2. Assume the link between C and X fails.
3. Using the distance vector routing algorithm, show the updates in routing tables for nodes A, B, and C for at least 3 iterations.
4. Demonstrate how the routing costs increase gradually, leading to three-node instability.
5. Briefly explain why this problem is more severe than two-node instability.